Report on Multiset and Prime Factors Programs (Haoran Ma)

Multiset Implementation

The Multiset class is a self-balancing binary search tree supporting duplicate elements. Each node stores a key and count, allowing efficient operations:

- Insert/Remove: Adds or deletes elements, adjusting counts.
- Contains/Count: Checks existence and frequency.
- Min/Max: Retrieves the smallest and largest elements.
- Floor/Ceil: Finds nearest elements based on a given key.

These operations run in O(log N) on average.

To verify functionality, a Google Test suite was expanded, adding:

- Duplicate Handling: Ensuring correct element counting.
- Exception Testing: Checking Count and Remove failures.
- Boundary Cases: Verifying Min, Max, Floor, and Ceil on edge conditions.
- Additional Test Cases: New tests were added to validate multiple duplicate insertions, verify the correctness of Floor and Ceil for both exact and non-exact matches, and test exception handling when querying or removing nonexistent elements.

Prime Factors Program

This program factorizes a number, storing prime factors in a MyMultiset, implemented using a sorted vector. It provides:

- Efficient Insertion and Lookup.
- Min/Max Retrieval.

Supported commands:

- all: Lists all factors.
- min/max: Retrieves smallest/largest factors.
- near: Finds closest factor.